

Solution Hints

1. Minimize Z

Subject to:

$$Z \geq x_1 - x_2 + 3x_3$$

$$Z \geq -x_1 + x_2 - 3x_3$$

$$Z \geq -x_1 + 3x_2 - x_3$$

$$Z \geq -x_1 - 3x_2 + x_3$$

$$x_1, x_2, x_3, Z \geq 0$$

2. $k = 0$; the solution is unbounded
 $k = 0.5$; optimal solution at (2,3) and (8,0)
 $k \geq 1$; optimal solution at (2,3)

Correct Ans: $k \geq 1$

3. $Z = -14$ at (0,15.33,10)